

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Botany**Paper: BOB-502 — Comparative Studies of Phanerogams**

Time: Three hours

Max. Marks: 70

Note: Attempt any five questions selecting two questions from each Section A and B. Question No. 9 (Section-C) is compulsory. The figures in the right-hand margin indicate marks.

SECTION – A

Question 1. Give a detailed account of the order Gnetales with special reference to *Gnetum*.
[15]

Question 2. Describe briefly any two of the following: [7.5 × 2 = 15]

- (a) Salient features of Ginkgoales
- (b) *Williamsonia*
- (c) Cycadales

Question 3. Give a brief account of the economic importance of gymnosperms. [15]

Question 4. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) *Pentoxylon*
- (b) Sexual reproduction in *Biota*
- (c) Embryogeny in conifers

SECTION – B

Question 5. With the help of floral formula and floral diagram describe the family Rutaceae. Give names of three economically important plants of the family. [15]

Question 6. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Outline of Takhtajan System of classification
- (b) Apomixis
- (c) Chemotaxonomy

Question 7. Write an account on embryo culture. Give its practical application. [15]

Question 8. Differentiate between any three of the following pairs of families: [5 × 3 = 15]

- (a) Asteraceae and Meliaceae
- (b) Euphorbiaceae and Moraceae
- (c) Liliaceae and Polygonaceae
- (d) Convolvulaceae and Scrophulariaceae

SECTION – C

Question 9. Answer the following in not more than fifty words each: [1 × 10 = 10]

- (a) Name an angiosperm-like character of *Gnetum*.
- (b) Define Pycnoxylic wood.
- (c) Define hard wood.
- (d) What is a bract scale?
- (e) Name a gymnosperm which is known as a living fossil.
- (f) Name a plant in which ochreate stipule is found.
- (g) Which type of placentation is found in Asteraceae?
- (h) Define adventive embryony.
- (i) Give the botanical name of a parasitic angiosperm.
- (j) Give the botanical name of a medicinal plant of Convolvulaceae.

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